Mahdi Salem,

Senior Data Engineer and Data Analyst Developer

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Turning operational complexity and data into decision intelligence.

Professional Profile

Data Engineer and BI Developer with 10+ years of experience building data integration, reporting, and automation solutions, including eight years supporting aerospace manufacturing at Bombardier. Experienced developing and maintaining enterprise data pipelines and Analytics engineering through ELT and SSIS packages development across multi-factory operations, integrating operational data, and establishing data-quality measures that help teams work from consistent information. Hands-on experience with Microsoft Fabric and Databricks, including Kafka, dbt, and PySpark in those environments. Combines practical engineering with business analysis and team leadership to make operational data more trusted, accessible, and useful for decision-making.

At Bombardier, I worked across legacy on-premises data warehouses—including Oracle 10g GBS—and modern data lake and lakehouse platforms. I supported the ingestion of SAP S/4HANA data into Azure Synapse and Databricks, and its progression and connection into Microsoft Fabric for BI analytical usage.

Core Competencies

- **Data integration and pipelines:** SSIS, ETL/ELT, data pipeline across operational systems, On-Prom & OneLake
- **Modern data platforms:** Microsoft Fabric and Databricks; Kafka, dbt, and PySpark in those environments
- **Data quality and reconciliation:** KPI frameworks, SAP/MES/local data reconciliation, data integrity, shared definitions
- **SQL and analytics:** SQL Server, T-SQL, PL/SQL, Python, Power BI, DAX, Power Query, semantic models
- **Operational data solutions:** Manufacturing, production, quality, workforce, attendance, and PMO reporting
- **Data governance and security:** RLS/OLS, data masking, master data and metadata practices
- **Engineering leadership:** Cross-functional delivery, team mentoring, stakeholder requirements, reusable standards
- **Process automation:** Azure Logical App, Power Automate, PowerApps, VBA, Python, and web applications

Professional Experience

Lead Data Engineer & BI Developer **Since Aug 2019**

Bombardier Aerospace *Mississauga, ON*

- Designed, developed, and maintained enterprise-grade SSIS packages across a multi-factory, multi-country footprint, improving data-flow accuracy and pipeline efficiency between core operational systems.
- Transmigrated new SAP S4/HANA database to Azure Databricks and lots of other sources to MS Fabric for data ingestion and Fabric capacity Optimization.
- Engineered and deployed over 150 Power BI reports, dashboards, and PowerApps solutions supporting Production Andon teams, Quality Assurance, and PMO governance, including 10+ standalone PowerApps built for fast, low-code process automation.
- Initiated and implemented a Data Quality KPI framework reconciling SAP, MES, and local production databases, materially improving shop-floor data integrity.
- Integrated Azure Data Lake and Azure Portal services to monitor and optimize Power BI Premium capacity usage, reducing resource overhead while maintaining query performance.

- Designed and launched an adaptive OSW (Outstanding Work) Production Tracking web application (.NET / Angular) to structure highly volatile, non-standard assembly tasks.
- Engineered a web-based attendance system with an administrator/supervisor dashboard using Python, Django, JavaScript, jQuery, and SQL for the OSW team.
- Created standardized report templates and mock-up frameworks under the company-wide Bombardier Operational Excellence (BOE) initiative, establishing best practices adopted across future reporting rollouts.
- Built 10+ rapid-response VBA applications automating cross-departmental communication and eliminating manual data entry.
- Led a team of up to 4 developers and data engineers through detailed planning and role clarity; coached and mentored team members, tailoring guidance and development opportunities to help each person contribute their best work.

Senior Data and Process Analyst **Jul 2018 – Aug 2019**

Triumph Aerospace Group *Toronto, ON*

- Engineered a custom VBA application to automate Crew Load Chart generation, cutting processing time from roughly 3 hours to 2 minutes and eliminating manual entry errors.
- Built a Python-based web application for daily attendance management with integrated Crew Load analytics to measure shift efficiency and optimize labor allocation.
- Architected a multi-tiered reporting framework covering daily, weekly, and monthly operational metrics at both macro and aircraft-structural levels.
- Developed real-time alerting mechanisms and dashboards delivering threshold notifications to line and executive managers to prevent project delays.

Process Improvement Lead **Aug 2016 – Jun 2018**

AAA Canada — Aerospace & Transportation *Toronto, ON*

- Built a Manpower Mapping application distinguishing touch vs. non-touch labor, improving shop-floor visibility and reducing non-value-added idle time.
- Architected a Line Logistic Cage tracking system giving real-time status on job kits and workability post-punch-out, preventing downstream assembly bottlenecks.
- Developed an automated Rapid Non-Conformance (RNC) reporting tool that cut generation time to 3 minutes, accelerating quality-containment response.
- Digitized Daily Management Meeting (DMM) reporting into an automated dashboard, lifting reporting efficiency by 200% while removing manual data entry.

Lead Business Consultant **Mar 2001 – Jun 2015**

Independent — Various Companies *Tehran, Iran*

- Directed 30+ consulting engagements in strategic planning, business process re-engineering, TQM and Six Sigma across multiple industries.
- Prepared 100+ proposals and RFQs; supported clients with performance and organizational improvement.

Selected Research & Advisory Experience

- Led market, feasibility, and investment studies across pharmaceutical, real-estate, retail, banking, public-sector, and industrial environments.
- Delivered business plans and feasibility studies used for investment decisions, financing applications, product selection, and new-business launches.
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Technical Expertise

BI & Analytics	Power BI (Direct Lake, Copilot, Semantic Models, RLS/OLS), DAX, Power Query (M), Tableau, SAS, Cognos, Azure Synapse
Data Engineering	SSIS, Microsoft Fabric (OneLake, Lakehouse, Data Factory), Databricks (Delta Lake, Unity Catalog, PySpark), Kafka, dbt, ETL/ELT & Medallion Architecture, Kimball Dimensional Modeling, Data Vault 2.0, 3NF, Schema-for-Change (SCD Type 1/2/3/6)
Platforms & Infrastructure	PostgreSQL, SQL Server (SSMS/SSIS, RLS/CLS), MySQL, Oracle (10g/12c, VPD), Azure Cosmos DB, Linux/Unix Administration, Docker, Cloud Networks, Relational & Semi-Structured Data (JSON/REST APIs)
Backend & Data Science	Python (Django, Flask, Pandas, NumPy, SciPy, scikit-learn, Matplotlib, openpyxl), PySpark, T-SQL, Jinja2, KQL, R, Node.js/Express, ASP.NET Core
UI / UX - Frontend	React, Angular, TypeScript, HTML5/CSS3, Bootstrap, jQuery, D3.js, Lodash
Low-Code / Power Platform	PowerApps, Dynamics 365, Dataiku, Power Automate, Power BI Embedded, CMS Frameworks (WordPress, Joomla, Drupal)
DevOps & Governance	Microsoft Purview, SQL Data Masking (DDM), Git, GitHub, Azure DevOps Pipelines, Bash, PowerShell

Education & Training

Master of Business Management, Marketing

Allameh Tabatabaie University, 1999 – 2002

Bachelor's Degree, Business Management

Allameh Tabatabaie University, 1996 – 1999

Professional Training

- Business Analytics; Duke University, 2016 – 2017
- CS50's Web Programming with Python and JavaScript; Harvard University, 2020
- Microsoft Power Platform (Power BI, PowerApps, Power Automate); 2020 – 2021

Certifications

- Data Engineering in Fabric Associate (DP-700)
- Fabric Analytics Engineer Associate (DP-600)
- Power BI Data Analyst Associate (PL-300)
- Power Platform App Maker Associate (PL-100)
- Power Platform Fundamentals (PL-900)
- Azure Data Fundamentals (DP-900)
- Excel 2016 Expert
- Certified ScrumMaster I (CSM I)

Six Microsoft badges held in total; some pending renewal.